STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 10: Thermal Properties of Matter

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

No net heat transfer between systems in thermal equilibrium.

300.15 K, approximately 300 K.

Heat needed to raise 1 kg by 1 K.

Q=mcDeltaT.

Q=mL.

Radiation.

pV=nRT.

Rate is approximately proportional to excess temperature for small differences.

4000 J.

1.2x10^-3 m.

373.15 K.

400 J/K.

66,800 J.

Energy supplies latent heat for phase change.

21,000 J.

1.2x10^-3 m.

44 deg C.

150 kPa.

Conduction via microscopic interactions; convection by bulk flow; radiation via electromagnetic waves.

mL+mcDeltaT=83,600 J.